frontiers | Frontiers in Climate

TYPE Original Research
PUBLISHED 26 January 2026
DOI 10.3389/fclim.2025.1719715

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OPEN ACCESS

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RECEIVED 06 October 2025
REVISED 16 December 2025
ACCEPTED 23 December 2025
PUBLISHED 26 January 2026

CITATION
Maniraho AP, Lanhai L, Bai J, Edwin IE, Nzabarinda V, Fabien M, Izere D, Niyomugabo P, Umugwaneza A and Bakayisire F (2026) Spatiotemporal analysis of agricultural drought and its relationship with climate variabilities in the growing season of the Horn of Africa.
Front. Clim. 7:1719715.
doi: 10.3389/fclim.2025.1719715

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Spatiotemporal analysis of agricultural drought and its relationship with climate variabilities in the growing season of the Horn of Africa

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This study investigates agricultural drought dynamics in the Horn of Africa (HoA) during the June–August growing seasons from 1995 to 2020. Using MODIS and GIMMS NDVI3g data, we derived the Vegetation Condition Index (VCI) to quantify patterns and identify climatic drivers. To delineate persistent drought vulnerability hotspots, we integrated temperature, precipitation, and VCI data, supplemented by satellite-derived soil moisture data, with a season-specific Weighted Overlay Index (WOI). Results revealed significant interannual variability in growing-season conditions, with widespread greenness in 1997–1998 contrasting with severe droughts in 2004, 2009, and 2011, the latter affecting up to 27.3% of the region. Drought hotspots were identified across northern Kenya, eastern Ethiopia, and central Somalia. Trend analysis indicates an increase in extreme and severe drought from 1995 to 2011, followed by partial contraction, though baseline drought exposure remains elevated. Climate data reveals significant spatial variability in precipitation and temperature, with a strong positive correlation between VCI and precipitation (R^2 = 0.73) and a negative correlation with temperature (R^2 = −0.74). Vulnerability assessment highlights western Ethiopia as highly vulnerable. Correlation analysis between VCI and detrended maize yield showed positive relationships, strongest in Ethiopia (R = 0.353) and Kenya (R = 0.415), validating VCI as an indicator of drought impact on crop performance. The findings underscore increasing drought frequency and the need for climate-resilient agricultural planning in the HoA.

KEYWORDS

agricultural drought, climate variability, Horn of Africa, spatiotemporal analysis, Vegetation Condition Index (VCI), Weighted Overlay Index (WOI)

1 Introduction

Drought poses a significant and growing threat to global food security and ecological stability, particularly in regions reliant on rain-fed agriculture (IPCC, 2021; World Bank, 2023). The Horn of Africa (HoA), characterized by its diverse topography, high climate variability, and heavy dependence on rain-fed agriculture, is exceptionally vulnerable to the impacts of drought (Nhemachena et al., 2020; Lottering et al., 2021).

Recurrent droughts in in this region lead to widespread crop failure, livestock losses, food insecurity, and socio-economic instability (Olagunju, 2015; Bayable and Gashaw, 2021; Maniraho et al., 2023). Understanding the complex interplay between climate variability and agricultural drought is therefore crucial for developing effective mitigation and adaptation strategies. Droughts are typically classified as meteorological, hydrological, or agricultural, each impacting different components of the water cycle and human systems (Van Loon, 2015; Wang et al., 2016). Agricultural drought specifically refers to periods of deficient soil moisture that negatively affects crop growth and yield (Begna, 2020; Dietz et al., 2021). Effective monitoring of agricultural drought is essential for risk reduction and climate-resilient planning (Nhemachena et al., 2020; Lottering et al., 2021). Traditional ground-based monitoring approaches, which rely on data from temperature, evaporation, and soil moisture sensors, often face limitations such as data scarcity, high operational costs, and spatial discontinuities, particularly in data-sparse regions like the HoA (AghaKouchak et al., 2015; Garg and Eslamian, 2017).

Satellite remote sensing offers a cost-effective and spatially consistent alternative for monitoring land surface dynamics over large areas, thereby overcoming many limitations of traditional methods (AghaKouchak et al., 2015; West et al., 2019; Jiao et al., 2021). Remote sensing-based drought indices have proven valuable for characterizing the spatiotemporal variability of agricultural drought (Measho et al., 2019; West et al., 2019). Among the various indices, the Vegetation Condition Index (VCI) is particularly suitable for heterogeneous landscapes. It normalizes the Normalized Difference Vegetation Index (NDVI) to its local seasonal range, enhancing sensitivity to anomalous vegetation stress by accounting for cumulative effects of rainfall, soil moisture, and management practices (Kogan, 1995a,b; Zeng et al., 2022; Wei et al., 2024).

The justification for using VCI lies in its ability to provide a more accurate reflection of vegetation health under diverse climatic conditions, essential for assessing drought impacts effectively (Dikici and Aksel, 2021; Rahman et al., 2025). While research on agricultural drought using remote sensing has expanded significantly, gaps remain regarding the nuances of drought dynamics within specific agro-ecological zones of the HoA. Previous studies, such as that by Gebrehiwot et al. (2016), utilized SPOT-VGT NDVI-based VCI (1998–2013) to map seasonal agricultural drought in Ethiopia. However, these studies often lack comprehensive assessments of drought attributes (frequency, intensity, persistence) aligned with agricultural calendars and detailed linkages to sub-regional climate patterns. This study innovatively integrates VCI with GIMMS NDVI3g and MODIS data to conduct a comprehensive analysis of agricultural drought dynamics and their relationship with climate variability in the HoA from 1995 to 2020. Building upon this foundation, the present study aims to provide a refined characterization of agricultural drought impacts across spatial and temporal scales in the HoA by focusing on growing season dynamics, integrating multi-metric drought attributes, and establishing sub-regional climate linkages. MODIS data, resampled to match the GIMMS NDVI3g resolution, extends the analysis beyond the original dataset's availability. The objectives of this study are as follows: (i) to quantify the spatiotemporal variability of agricultural drought during the growing season; (ii) to evaluate the frequency and spatial patterns of agricultural drought trends; (iii) to assess spatiotemporal changes in climate variability and their implications for agricultural drought characteristics; and (iv) to map agricultural drought vulnerability hotspots using a Weighted Overlay Index, integrating soil moisture, VCI, precipitation, and temperature.

In this way, our approach advances the existing literature by aligning analysis with the agricultural calendar and isolating growing season dynamics over a multi-decadal period. We quantify and map the frequency, intensity, and persistence of drought, generating a comprehensive risk profile that is novel in the context of the Horn of Africa.

Furthermore, we elucidate how VCI-based drought metrics relate to sub-regional climate variability, clarifying where and when climate anomalies translate into vegetation stress across heterogeneous landscapes. By delineating drought vulnerability zones through a multi-indicator weighted overlay, we produced an actionable map to guide preparedness, early warning, and climate-resilient planning. The findings support the prioritization of interventions such as drought-tolerant varieties and climate-informed planting calendars, thereby strengthening early warning systems that reflect the timing and persistence of drought stress in resource-limited, high-stakes settings. Anticipating the recurrence, severity, and duration of drought is essential for protecting livelihoods and sustaining agricultural productivity in the Horn of Africa.

2 Materials and methods

2.1 Description of the study area

The study focuses on the Horn of Africa (HoA), which includes Kenya, Ethiopia, Eritrea, Somalia, and Djibouti (Figure 1), and covers approximately 5.9 million km^2. This region exhibits pronounced heterogeneity in topography, rainfall regimes, and vegetation. Climate varies from humid highlands in parts of Kenya and Ethiopia to arid and semi-arid lands (ASALs) across much of the lowlands.

Most equatorial areas experience bimodal rainfall, characterized by "long rains" from March to May (MAM) and the "short rains" from October to November (OND); northern areas generally have a single primary rainy season in June–August (JJA) (Nicholson, 2017). Annual rainfall variability is strongly influenced by the meridional migration of the Intertropical Convergence Zone (ITCZ), while local to regional rainfall distributions are shaped by factors such as topography, proximity to coastlines, monsoon dynamics, and remote climate modes including the El Niño–Southern Oscillation and the Indian Ocean Dipole (Nicholson, 2017; Camberlin, 2018).

The study emphasizes the growing season for agricultural drought analysis, which varies by country and crop type. A detailed overview of crop calendars for each country is provided in supplementary material, but key growing periods can be summarized as follows: In Eritrea, most crops are planted from June–July, with growth primarily

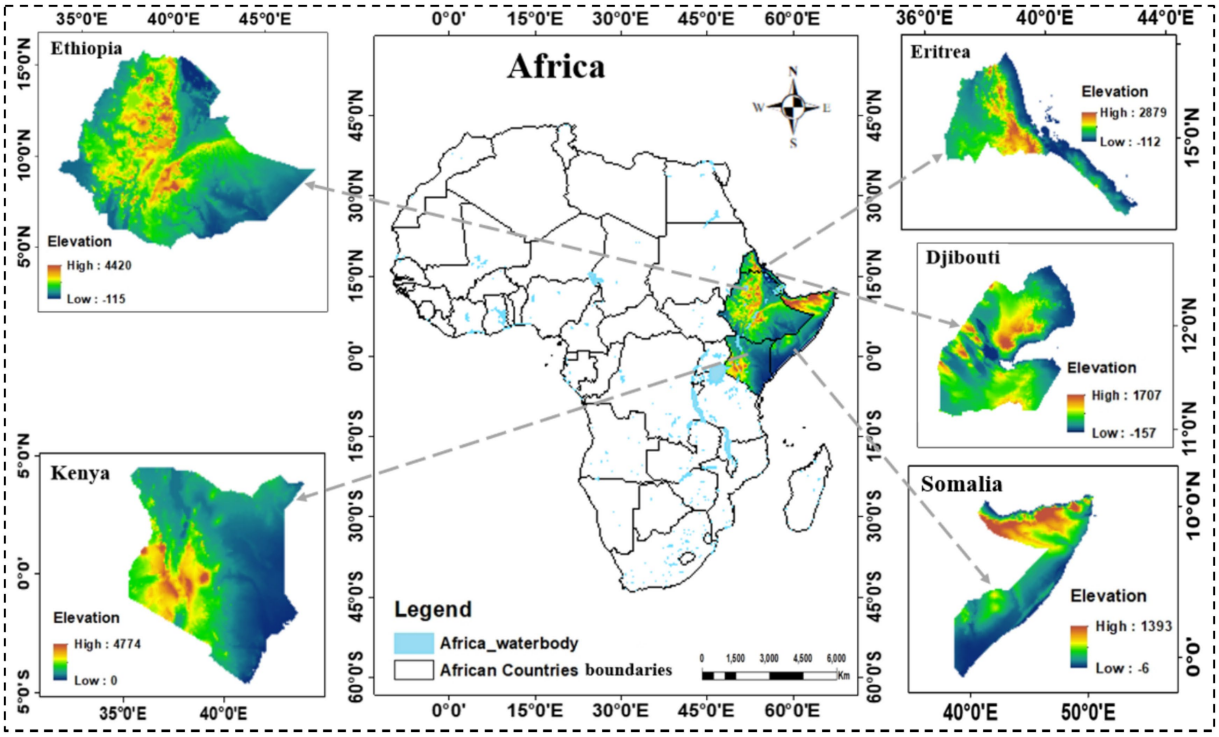


FIGURE 1
The location of the study area shows digital elevation model maps of different countries as well as the African map with country boundaries.

August–October. In Kenya, planting typically occurs March–October, yielding a growing period April–November. In Djibouti, planting is concentrated June–October, with growth July–November. In Ethiopia, planting is more diverse, commonly February–August, with growth March–October. In Somalia, most crops (excluding rice) are planted in April, with the main growth season May–July. These patterns underscore marked subregional differences in planting and growth windows across the HoA.

This study concentrates on June–August (JJA) because it aligns with the dominant main season across large parts of the HoA, captures peak vegetation activity and water stress, and maximizes the temporal consistency of satellite observations for comparative analysis.

2.2 Sources of data

2.2.1 VCI data

The Vegetation Condition Index (VCI), a standardized indicator of vegetation status relative to historical conditions (Liang et al., 2017), was used to assess drought extent during the growing season. VCI values were derived from seasonal mean Normalized Difference Vegetation Index (NDVI) values calculated from June to August for each year from 1995 to 2020. To compute these seasonal NDVI values, we primarily utilized the Global Inventory Monitoring and Modeling System (GIMMS) NDVI3g dataset (available since 1981, spatial resolution of 0.083°, 15-day temporal resolution; https://gimms.gsfc.nasa.gov). For years beyond the availability of GIMMS NDVI3g, MODIS data was used, resampled to match the GIMMS NDVI3g resolution. These seasonal NDVI values were then converted into seasonal VCI time series for drought trend analysis across the study area. It is important to acknowledge several limitations associated with both the GIMMS NDVI3g and the resampled MODIS datasets when interpreting results. The relatively coarse spatial resolution may mask drought signals, particularly in smaller or heterogeneous agricultural landscapes. Cloud-induced gaps can degrade the reliability of vegetation assessment during critical drought periods. Furthermore, the 15-day revisit interval may miss rapid vegetation changes, potentially limiting the detection of short-term drought events. These limitations warrant cautious interpretation of the VCI results and suggest that future research could benefit from exploring complementary data sources to further strengthen the robustness of findings.

2.2.2 Climate data

Two climate datasets were used in this study: CHIRPS for precipitation and ERA5 for temperature. The study period spans 1995–2020, and both datasets provide continuous, high-quality records within this timeframe.

The Climate Hazards Group Infrared Precipitation with Stations (CHIRPS) dataset provides global rainfall estimates between 50°S and 50°N from 1981 to the present.[1] CHIRPS combines satellite-derived imagery with in-situ station data at a spatial resolution of 0.05° (~5 km) to produce long-term, gridded precipitation time series suitable for hydrological and climate applications.

1 https://data.chc.ucsb.edu/products/CHIRPS/

The ERA5 reanalysis dataset, produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) under the Copernicus Climate Change Service, offers a wide range of atmospheric and surface variables, including 2-m air temperature. ERA5 provides global coverage from 1979 to the present at an hourly temporal resolution and a spatial resolution of 0.25° (~31 km). The dataset was accessed through the Copernicus Climate Data Store.[2] CHIRPS was selected due to its extensive spatial and temporal coverage and its demonstrated reliability for rainfall monitoring and drought assessments, particularly across Africa. Previous evaluations have shown that CHIRPS exhibits lower systematic and mean absolute errors compared with other global precipitation products such as the GPCC (Maniraho et al., n.d.; Yang et al., 2024), making it well suited for regional-scale climate and hydrological analysis (Dutta, 2018; Habitou et al., 2020; Sacré Regis et al., 2020). ERA5 was chosen because it represents the most recent and advanced ECMWF reanalysis product, offering improved spatial and temporal resolution, enhanced data assimilation, and better representation of temperature variability compared to its predecessor ERA-Interim (Soci et al., 2024; Zhang Y. et al., 2024). The use of ERA5 ensures data consistency and continuity throughout the 1995–2020 study period (Lombe et al., 2024). Since ERA5 (for temperature) and CHIRPS (for precipitation) represent distinct climatological variables derived from independent sources, their combined use does not affect temporal or spatial continuity in the analysis.

2.2.3 Soil moisture

This study used gridded surface soil moisture from the European Space Agency Climate Change Initiative (ESA CCI) Soil Moisture product (combined active–passive), which provides daily top-layer estimates (0–5 cm) at 0.25° resolution from 1978 onward (Liu, Li et al., 2023), We focused on the Horn of Africa, restricted to the 1995–2020 period and the growing season (June–August). Daily values were aggregated to monthly and then seasonal (JJA) means for each year. To enable cross-variable comparability, soil moisture, precipitation, and temperature were standardized per grid cell using z-scores relative to the 1995–2020 baseline. These standardized anomalies, together with the Vegetation Condition Index (VCI), were integrated to delineate areas most vulnerable to agricultural drought. ESA CCI quality flags were applied, and grid cells with insufficient temporal coverage were masked.

2.2.4 Crop yield data

Maize yield data for countries in the Horn of Africa were sourced from the FAOSTAT online platform (https://www.fao.org/faostat/en/#data/QC, accessed January 1, 2020), a database maintained by the Food and Agriculture Organization of the United Nations. This dataset includes annual, national-level maize production statistics from 1995 to 2020. Because FAO yield data are based on aggregated national agricultural statistics, they represent average production conditions across the primary maize-growing areas within each country. This national-scale spatial resolution is appropriate for broad-scale drought assessments and analyses of year-to-year variability. To isolate the impact of climate and drought on yield, the maize yield data were detrended to eliminate the influence of factors such as technological advancements, changes in agricultural practices, and long-term productivity trends. This detrended yield data was then used to validate our findings by correlating it with the growing season Vegetation Condition Index (VCI) for each country. This correlation analysis allowed us to evaluate the relationship between vegetation health and maize production.

2.3 Methods

2.3.1 Calculations of VCI

To assess drought extent and intensity during the growing season, we used the Vegetation Condition Index (VCI), which measures vegetation health anomalies relative to long-term conditions derived from the Normalized Difference Vegetation Index (NDVI). Two NDVI sources were used: GIMMS NDVI3g and MODIS NDVI. To ensure spatial consistency between these datasets, MODIS NDVI was resampled to match the GIMMS NDVI3g resolution. This resampling allowed seamless integration, reducing spatial-scale bias and enabling consistent pixel-by-pixel VCI comparisons across the time series.

The VCI was computed using the following Equation 1 (Mutie, 2023):

$$\mathrm{VCI} = \left(\frac{\mathrm{NDVI}_i - \mathrm{NDVI}_{\min}}{\mathrm{NDVI}_{\max} - \mathrm{NDVI}_{\min}} \right) \times 100 \quad (1)$$

where NDVI_i represents the NDVI value for a specific month (June–July), while $\mathrm{NDVI}_{\min}$ and $\mathrm{NDVI}_{\max}$ are the minimum and maximum NDVI values for the same period over a 26-years (1995–2020).

VCI values range from 0% (extreme drought) to 100% (optimal vegetation condition). Drought severity classes were assigned as shown in Table 1, following (Kogan, 1995a,b).

2.3.2 The intensity of drought

Drought intensity was assessed for each year during the main growing season (June–August) by calculating the proportion of pixels (representing land surface area) within the VCI-defined drought severity classes (Đidelija et al., 2023). This step reveals interannual variations in drought severity and spatial progression across the study area.

2.3.3 The computation of drought frequency

The overall frequency of drought encompasses the cumulative frequencies of normal, mild, moderate, severe, and very severe drought. Based on the degree of overall drought frequency, 0.8 and

TABLE 1 Drought severity classes based on VCI.

Drought severity classes	VCI value
Extreme	$0 \leq \mathrm{VCI} < 10$
Severe	$10 \leq \mathrm{VCI} < 20$
Moderate	$20 \leq \mathrm{VCI} \leq 30$
Mild	$30 \leq \mathrm{VCI} \leq 40$
No drought	$40 < \mathrm{VCI} \leq 100$

2 https://cds.climate.copernicus.eu/

above is regarded as high frequency, while 0.4 and below is classified as low frequency (Qian et al., 2023). The calculation of the various degrees of drought frequency (*f*) was performed using the Equation 2 below, as adapted from (Qian et al., 2023).

$$f = \frac{n}{N} \tag{2}$$

where f represents the frequency of drought, n indicates the number of distinct drought categories during the study period, and N signifies the overall duration of the study period.

2.3.4 Mann—Kendall (MK) test

To examine temporal trends in climatic and vegetation variables, the Mann–Kendall (MK) test a non-parametric rank-based test robust to non-normal data distributions was used (Militino et al., 2020; Mohammad et al., 2022). The MK statistic (S) was computed as Equation 3.

$$S = \sum_{j<1}^{n} \mathrm{Sgn}\left(Z_i - Z_j\right), \tag{3}$$

where Z_i and Z_j represent average seasonal values of VCI, precipitation, or temperature for years (i) and (j), respectively, and Sgn ($Z_i - Z_j$) is defined as Equation 4.

$$\mathbf{Sgn}\left(Z_i - Z_j\right) = \begin{Bmatrix} \text{if}\left(Z_i - Z_j\right) < 0; & \text{then} & -1 \\ \text{if}\left(Z_i - Z_j\right) = 0; & \text{then} & 0 \\ \text{if}\left(Z_i - Z_j\right) > 0; & \text{then} & 1 \end{Bmatrix} \tag{4}$$

A positive Z-score indicates an upward trend while a negative Z-score signifies a downward trend. Statistical significance was assessed at the 95% confidence level (Zhang Z. et al., 2024).

2.3.5 Correlation between VCI and climate variables

The Pearson correlation coefficient (r) was applied to evaluate the linear relationship between mean seasonal VCI and climate variables (precipitation and temperature) across the 1995–2020 period. In this study, the correlation coefficients for each pixel were calculated using Equation 5:

$$r_{x,y} = \frac{\sum_{i=1}^{n}\left(x_i - \overline{x}\right)\left(y_i - \overline{y}\right)}{\sqrt{\sum_{i=1}^{n}\left(x_i - \overline{x}\right)^2 \sum_{i=i}^{n}\left(y_i - \overline{y}\right)^2}} \tag{5}$$

where r_{xy}, represents the correlation coefficient, n denotes the duration of the time series, and i indicates the number of the year from 1995 to 2020 (26 years). x_i and y_i represent VCI and climate variable values for year i, and $\overline{x}$ and $\overline{y}$ are their respective average. A positive relationship is indicated when $r_{xy} \geq 0$; a negative relationship when $r_{xy} \leq 0$; and it is indicated that there is no relationship when $r_{xy} = 0$ (Tong et al., 2017; Lu et al., 2023).

2.3.6 Delineation of agricultural drought vulnerability areas

To identify spatial patterns of agricultural drought vulnerability across the Horn of Africa, a Weighted Overlay Index (WOI) was developed by integrating four standardized biophysical including precipitation, temperature, Vegetation Condition Index (VCI), and satellite-derived soil moisture, averaged for the June–August growing season (1995–2020). Each variable was normalized to a common 1–5 vulnerability scale, where higher values denote greater drought susceptibility.

Weights for the individual indicators were assigned based on their relative importance in influencing agricultural drought, informed by existing literature and regional expert evaluation (Bera and Dutta, 2024; Senapati and Das, 2024).

Consistent with previous composite drought vulnerability frameworks, precipitation (0.35) and soil moisture (0.30) received the largest weights as primary determinants of crop water availability. VCI (0.25) was assigned an intermediate weight because it reflects vegetation response to both water and temperature stress, and temperature (0.10) was given a lower weight, representing its secondary effect through evapotranspiration and thermal stress.

The Weighted Overlay Index (WOI) was calculated using Equation 6 (Aprina et al., 2024):

$$\mathrm{WOI} = \sum_{i=1}^{n}\left(W_i \times X_i\right) \tag{6}$$

where W_i is the assigned weight of variable i, and X_i is the standardized vulnerability class (1–5) for that variable. The resulting continuous WOI values range between 0 and 1, with higher values indicating greater overall vulnerability. To translate the WOI into discrete vulnerability classes, the resulting raster was categorized using the Natural Breaks (Jenks) classification method, which identifies breaks in the data distribution to form internally homogeneous and externally distinct groups. Based on this, five vulnerability levels were defined: Very Low (0–0.2), Low (0.2–0.4), Moderate (0.4–0.6), High (0.6–0.8), and Very High (> 0.8). This classification approach ensures that the final vulnerability map captures the relative severity of drought conditions by combining the dominant environmental drivers (precipitation and soil moisture) with vegetation and temperature responses.

2.3.7 Validation of VCI using field-based maize yield data

To ensure the reliability and accuracy of the Vegetation Condition Index (VCI) and associated drought assessments, we validated the seasonal VCI against field-based maize yield data for the Horn of Africa over the period 1995–2020. Specifically, we used June–August (JJA) seasonal VCI values and correlated them with maize yield data obtained from local agricultural assessments for corresponding years and locations.

Prior to the correlation analysis, maize yields were detrended to remove long-term increases associated with technological advancement, improved management practices, and policy changes. Detrending is crucial because such non-climatic trends can obscure the true relationship between climate variability (as captured by VCI)

and crop performance. By working with detrended yield anomalies, the analysis more accurately reflects interannual climate-driven variability and its impact on maize production.

Correlation analyses between detrended maize yields and seasonal VCI (June–August, 1995–2020) were then conducted to evaluate how well VCI captures drought-related stress affecting agricultural outcomes. This integration of ground-based yield data provided a clearer understanding of how VCI variations correspond to actual agricultural performance in the Horn of Africa.

3 Results

3.1 The spatiotemporal evolution of agricultural drought conditions

The Analysis of the Vegetation Condition Index (VCI) derived from MODIS and GIMMS NDVI3g data revealed pronounced interannual variability in vegetation health across the HoA (Figure 2).

Generally favorable vegetation conditions prevailed, interrupted by moderate drought episodes in 1996 and 1999. The region experienced widespread greenness in 1997–1998, when more than 70% of the area showed "no drought" conditions, consistent with wet anomalies linked to the strong El Niño event.

A marked deterioration occurred, with multiple severe droughts, particularly in 2004, 2009, and 2011. Extreme drought affected 16.3% of the area in 2004 and peaked at 27.3% in 2011, the most extensive drought within the study period. Spatially, drought impacts expanded from northeastern Ethiopia and southern Somalia (2004) to southern Kenya and southwestern Ethiopia (2009), culminating in region-wide vegetation stress in 2011. These years align with well-documented food security crises (Moyo, 2024; Tesfaye et al., 2025).

After 2011, drought severity declined but remained higher than in the 1990s. Extreme drought covered 8.1% in 2016 and 5.2% in 2020, indicating modest recovery. However, this improvement did not fully compensate for earlier ecosystem impacts, as severe drought persisted over 6–12% of the region during several years (2016–2019). The 2019 episode mainly affected northeastern and southeastern Kenya and southern Somalia, demonstrating the shifting spatial footprint of drought.

The long-term trend reveals an increase in the area affected by extreme and severe drought from 1995 to 2011, followed by partial contraction afterward (Figure 3). Although the share of "no drought" conditions rose to 54.2% by 2020, it remained well below 1997 levels (76.7%), indicating higher baseline drought exposure. Spatial analysis highlights persistent hotspots of vulnerability across northern Kenya, eastern and northeastern Ethiopia, northwestern Eritrea, and central Somalia, where vegetation consistently showed low VCI values linked to below-average rainfall and warming trends.

Agricultural drought across the HoA exhibits a cyclical yet intensifying pattern: widespread wet conditions in the late 1990s were succeeded by severe, prolonged droughts in the late 2000s and early 2010s, followed by partial recovery without full ecosystem rebound. This evolution aligns with reports of increasing drought frequency and severity in East Africa, underscoring continuing challenges for climate adaptation and drought-resilient agriculture (Ukwuaba Jacintha, 2024; Tefera et al., 2025).

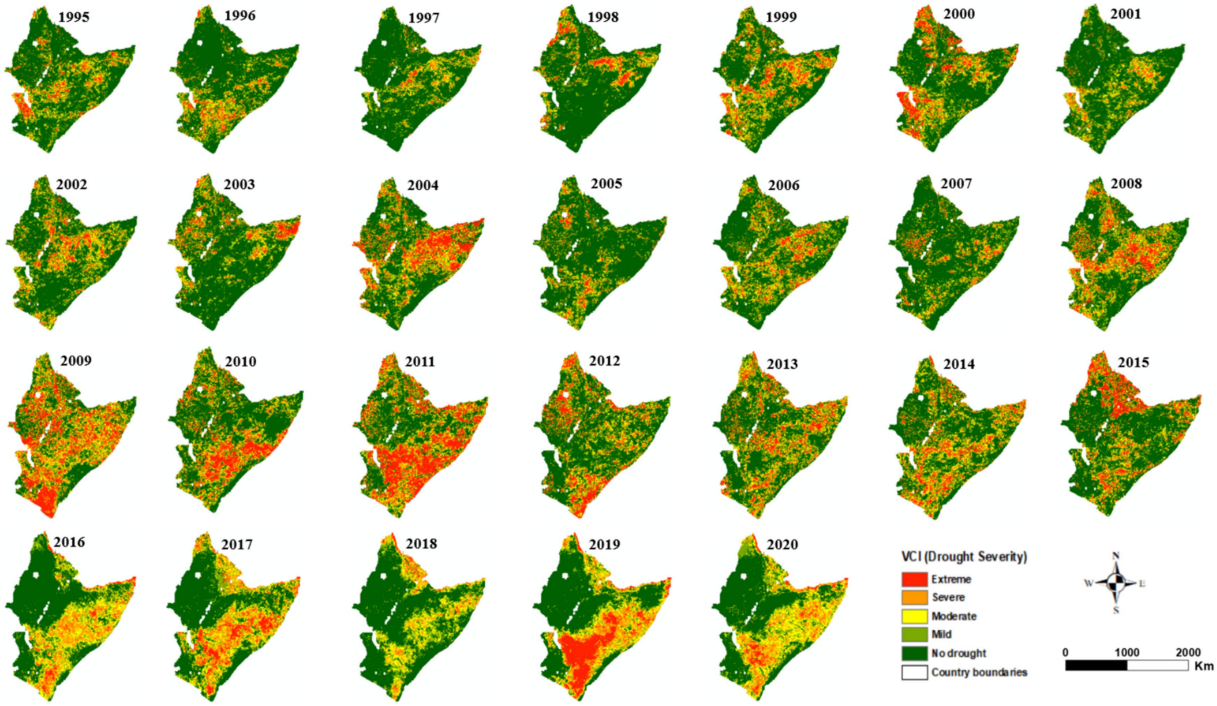


FIGURE 2
Spatial pattern of agricultural drought condition based on growing seasonal VCI in the HOA.

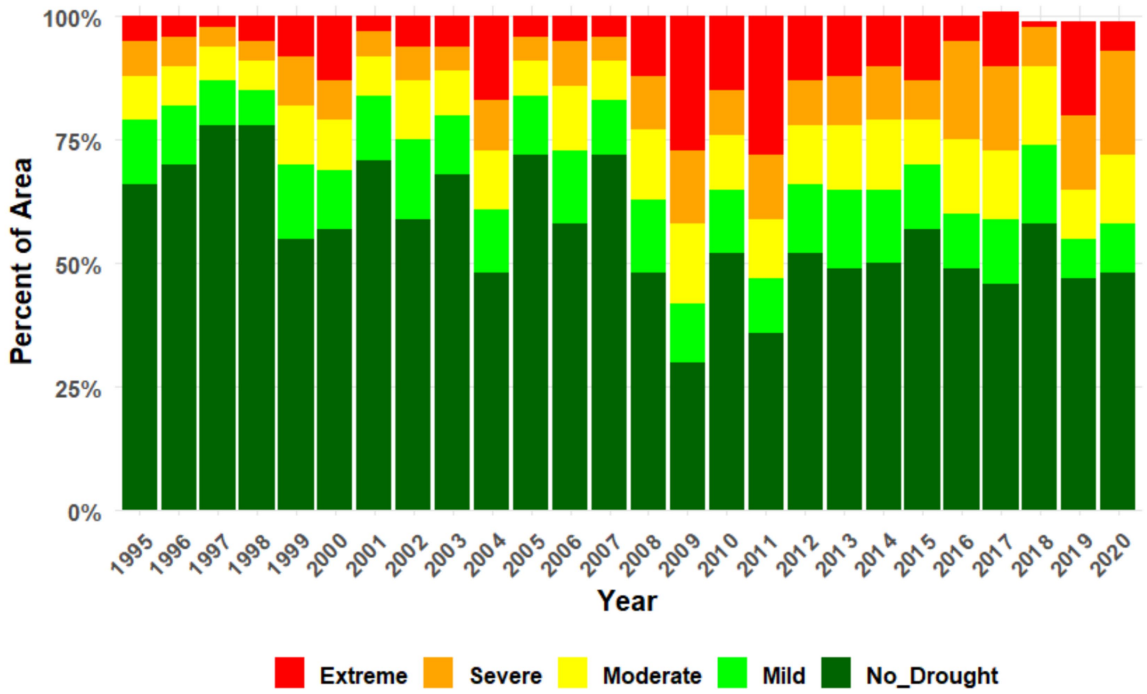


FIGURE 3
The percentage of land area influenced by different levels of drought and non-drought conditions during growth season from 1995 to 2020.

3.2 The distribution of agricultural drought frequency

The spatial pattern of agricultural drought frequency across the Horn of Africa (HoA) during the 1995–2020 growing seasons demonstrates substantial spatial heterogeneity (Figure 4). Drought occurred frequently across much of the region, with large areas exhibiting frequency values greater than 0.4. High-frequency zones (above 0.8) are concentrated in the eastern and northeastern parts of Ethiopia, central and southern Somalia, and northern Kenya, indicating regions where vegetation repeatedly experienced stress during the observation period.

In contrast, relatively lower drought frequencies (below 0.4) are evident in central and western Ethiopia, parts of Uganda, and along certain highland areas where vegetation conditions were more stable across seasons. The spatial gradients reflect the combined influence of regional climatic variability, terrain, and rainfall distribution patterns.

While drought events were common, most were of mild to moderate intensity, suggesting that recurrent vegetation stress, rather than isolated extreme events, dominates the regional drought regime. These findings emphasize the persistence of drought exposure in key agricultural zones and highlight the need for long-term adaptation and vegetation recovery planning rather than short-term crisis response alone.

3.3 Agricultural drought trend analysis

Trends in the Vegetation Condition Index (VCI) during the June–August growing seasons from 1995 to 2020, assessed using the Mann–Kendall test, reveal moderate spatial variability across the Horn of Africa (Figure 5). The regional mean shows a slight, statistically non-significant decreasing trend in VCI, indicating limited overall change in vegetation health during the study period. However, spatially distinct patterns emerge among individual countries.

In Ethiopia, significant decreasing trends are observed over eastern and central areas, corresponding to sustained vegetation decline. These patterns are consistent with earlier findings that highlight increasing drought frequency and land degradation in these regions (Abara and Budiastuti, 2020; Mohammed et al., 2022).

In contrast, Southwestern Ethiopia and parts of western Kenya show stable or weakly positive VCI trends, suggesting comparatively better vegetation conditions. Previous rainfall and vegetation studies associate these areas with relatively higher and more consistent rainfall during the main growing season (Camberlin, 2018).

Somalia exhibits predominantly non-significant declining trends, although localized stability is evident along the southern regions. In Kenya, notable decreases appear in the northern and eastern zones, while central highlands generally remain stable. Eritrea and Djibouti display mostly negative trends, aligning with their more arid climatic regimes and limited vegetation cover.

3.4 Spatial and temporal distribution climate variability from 1995 to 2020

The spatial distribution of climate variables across the Horn of Africa during the growing season from 1995 to 2020 reveals significant variability in both mean precipitation and temperature. As depicted in Figure 6a, mean precipitation across the region varied considerably, with a maximum recorded value of 392.7 mm. Areas in the southwestern part of Ethiopia received the highest precipitation, indicated in red, while much of Somalia and eastern Kenya experienced significantly lower rainfall levels, often recorded as near-zero millimeters. This disparity in precipitation is critical, as it directly influences agricultural

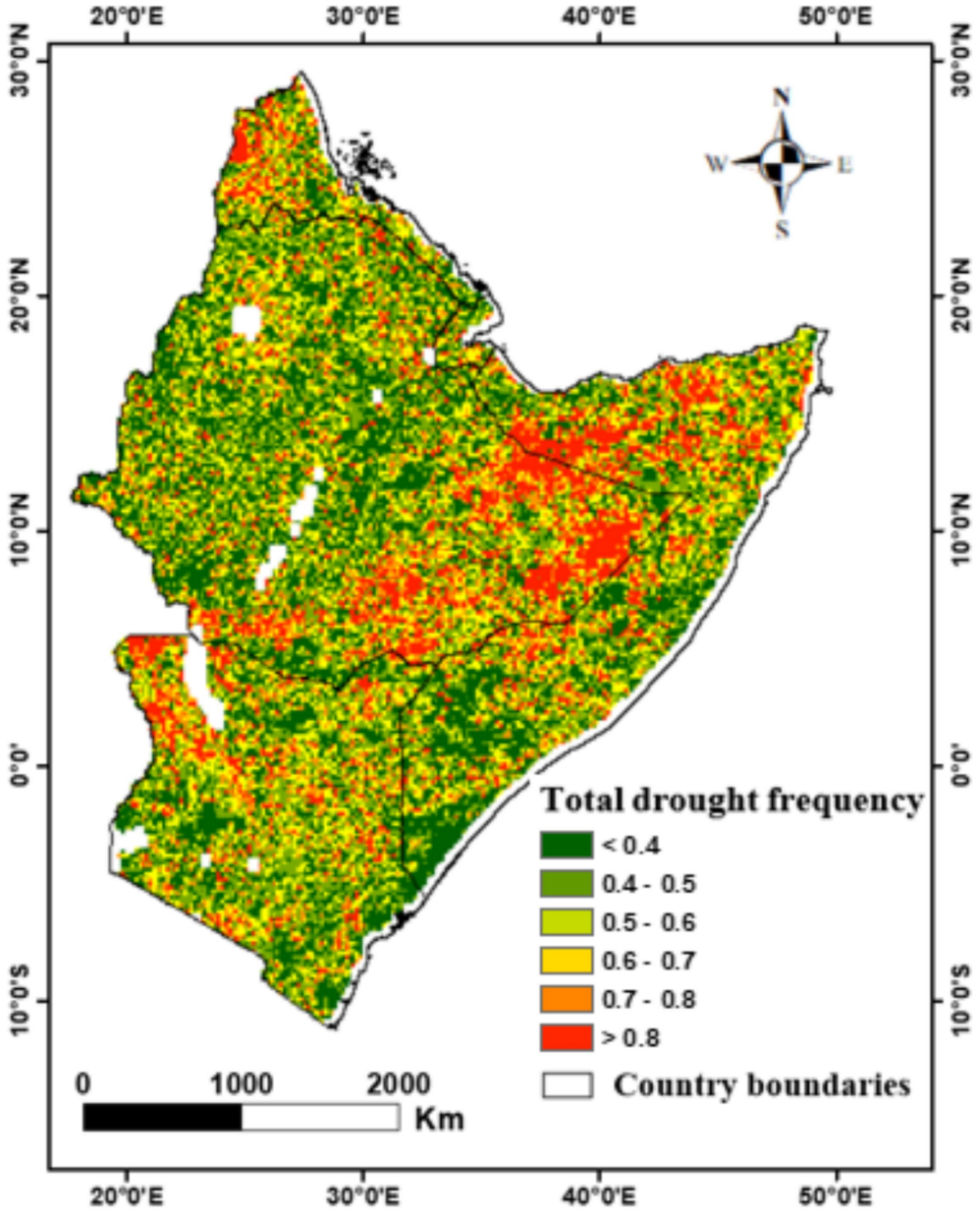


FIGURE 4
The spatial distribution of the total drought frequency during the growing season from 1995 to 2020.

productivity and water availability, which are essential for sustaining local livelihoods.

In contrast, Figure 6b presents the mean temperature distribution, with an average high of 36.3 °C across the region. Areas with the highest temperatures are predominantly located in eastern and northeastern Kenya, where values fluctuate around the upper limits of the temperature scale. These elevated temperatures, combined with low precipitation levels in certain areas, potentially exacerbate drought conditions, impacting vegetation health and promoting arid landscapes.

Additionally, Figure 7 illustrates seasonal precipitation and temperature trends from 1995 to 2020 across five East African countries. The region shows persistent spatial and temporal variability in climate. Djibouti and Eritrea experience consistently low precipitation and relatively high temperatures, reflecting their arid environments. Kenya and Somalia exhibit fluctuating rainfall with intermittent wet and dry periods, indicative of high interannual variability. In Ethiopia, precipitation displays an increasing tendency throughout the period, particularly in the southwestern regions, while temperature has steadily risen across all countries. These contrasting patterns highlight differing levels of hydro-climatic stress within the Horn of Africa.

3.5 Climate variability and its relationship with vegetation condition index

Figure 8 summarizes the statistical associations between the Vegetation Condition Index (VCI) and the two key climatic

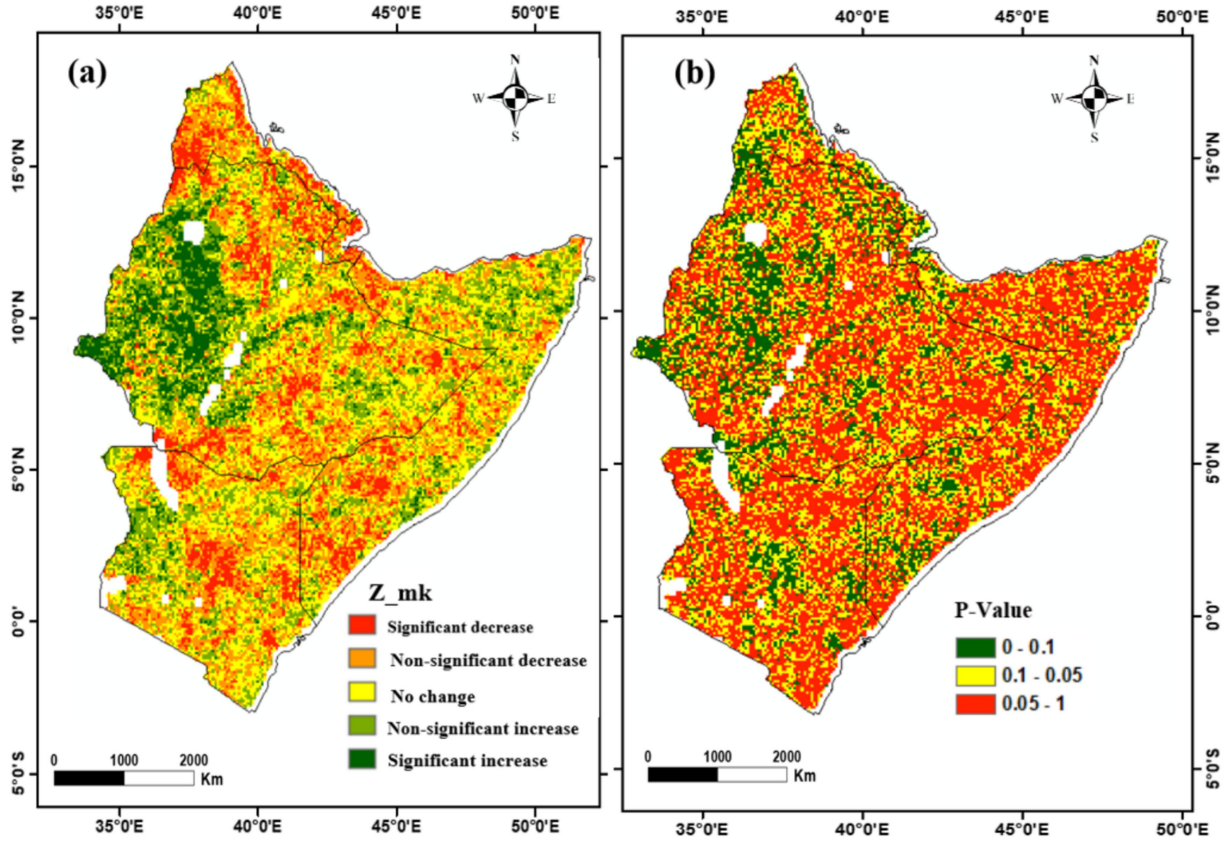


FIGURE 5
Trends in the growing season VCI in the Horn of Africa from 1995 to 2015: **(a)** VCI trend; **(b)** significance (*p*-value) of VCI trend.

variables during the growing season. A strong positive relationship is observed between VCI and precipitation ($R^2 = 0.73$) (Figure 8a), indicating that higher rainfall generally supports healthier vegetation conditions. In contrast, VCI exhibits a significant negative correlation with temperature ($R^2 = -0.74$) (Figure 8b), suggesting that elevated temperatures are often linked to vegetation stress. However, the relationship with temperature is not strictly linear warmer conditions may initially enhance plant growth up to a physiological threshold, beyond which moisture loss and evapotranspiration stress dominate, leading to VCI decline.

Together, these results confirm that vegetation health in the Horn of Africa is strongly influenced by the interplay between rainfall availability and thermal stress. The combined analysis underscores the need for climate-responsive agricultural planning that accounts for both precipitation variability and temperature-driven soil moisture deficits when assessing drought vulnerability.

Additionally, the study evaluated the correlation coefficients between VCI and precipitation across different countries in the Horn of Africa during the growing season, as illustrated in Figure 9. The results revealed varying strengths of positive correlations. Somalia exhibited a strong positive correlation ($R^2 = 0.86$) between VCI and precipitation, indicating a robust relationship where increased rainfall is associated with improved vegetation conditions. Kenya also demonstrated a significant correlation ($R^2 = 0.71$), suggesting that precipitation positively influences vegetation health in the region. Ethiopia had a similarly strong correlation ($R^2 = 0.68$), underscoring the significance of rainfall for enhancing vegetation condition.

In contrast, Eritrea showed a moderate correlation ($R^2 = 0.53$), while Djibouti exhibited a weaker correlation ($R^2 = 0.59$) compared to the other countries. These correlations indicate that while precipitation positively affects VCI across all evaluated countries, the strength of this relationship varies, reflecting differing climatic conditions and agricultural practices.

Furthermore, the analysis of correlations between VCI and temperature across the Horn of Africa revealed strong negative relationships, as illustrated in Figure 10. Notably, the negative correlations between growing season mean VCI and temperature were strongest in Kenya ($R^2 = 0.92$) and Somalia ($R^2 = 0.82$), indicating that lower temperatures are associated with improved vegetation conditions in these countries.

Eritrea also displayed a moderate negative correlation ($R^2 = 0.47$), while Djibouti showed a correlation of $R^2 = 0.59$. In contrast, Ethiopia exhibited a weaker correlation ($R^2 = 0.26$). These findings underscore the overall trend that lower temperatures during the growing season correspond to healthier vegetation, reinforcing the idea that elevated temperatures can have detrimental effects on vegetation health. This observation aligns with Abera et al. (2018), who highlighted the adverse impacts of high temperatures on vegetation vigor.

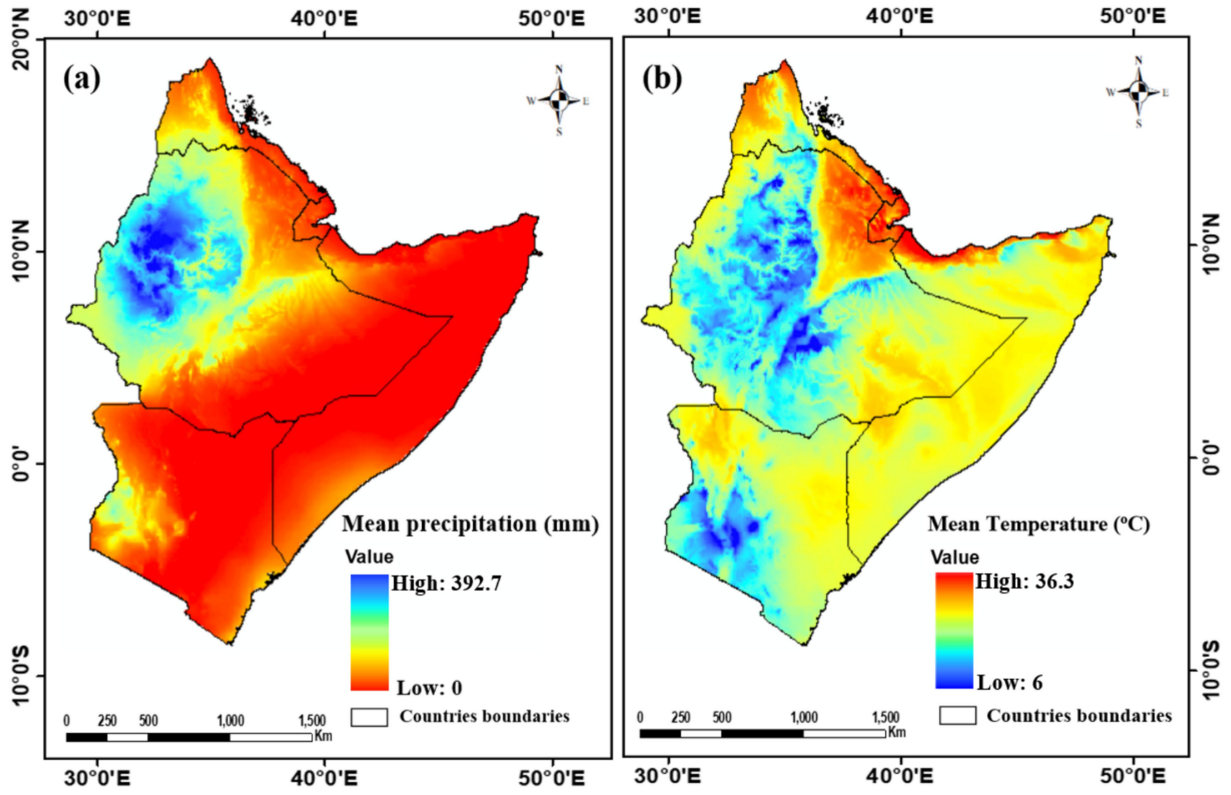


FIGURE 6
Spatial distribution of **(a)** mean precipitation in millimeters and **(b)** mean temperature in degrees Celsius across the Horn of Africa during the growing season, from 1995 to 2020.

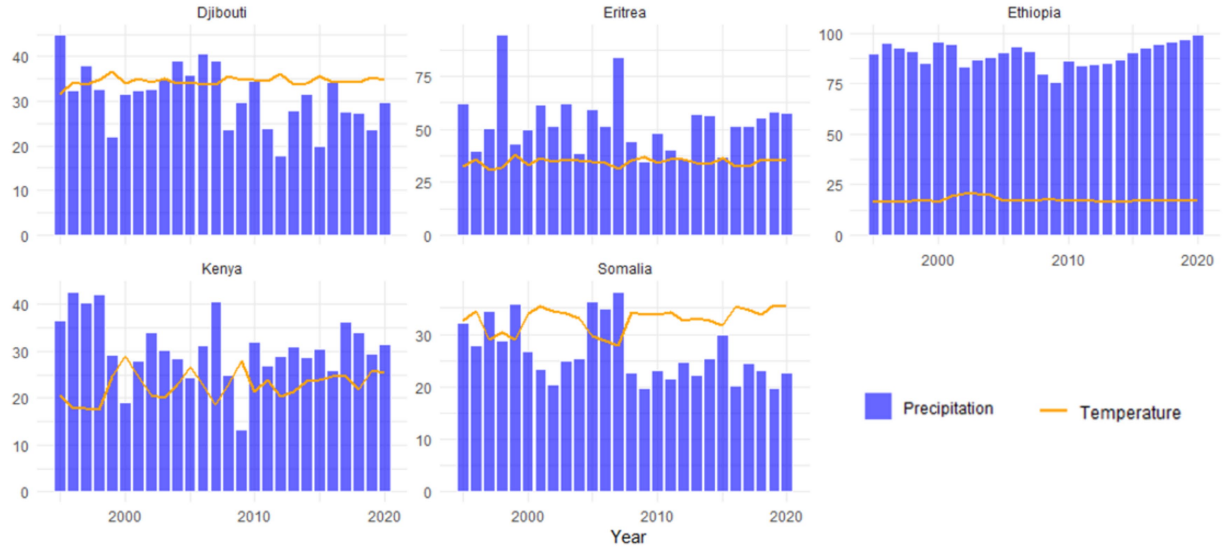


FIGURE 7
Trends in annual precipitation and temperature from 1995 to 2020 in different countries of the Horn of Africa.

3.6 Spatial distribution of agricultural drought vulnerability during growing season

The spatial distribution of vulnerability across the Horn of Africa displays pronounced regional contrasts (Figure 11a). Western and northwestern Ethiopia forms contiguous zones of high to very high vulnerability, extending toward Eritrea and the northern highlands. Moderate vulnerability dominates the central Ethiopian highlands, illustrating the gradient of drought risk within the country. In contrast, much of eastern Ethiopia and Somalia show low to moderate vulnerability, interspersed with localized high-risk pockets.

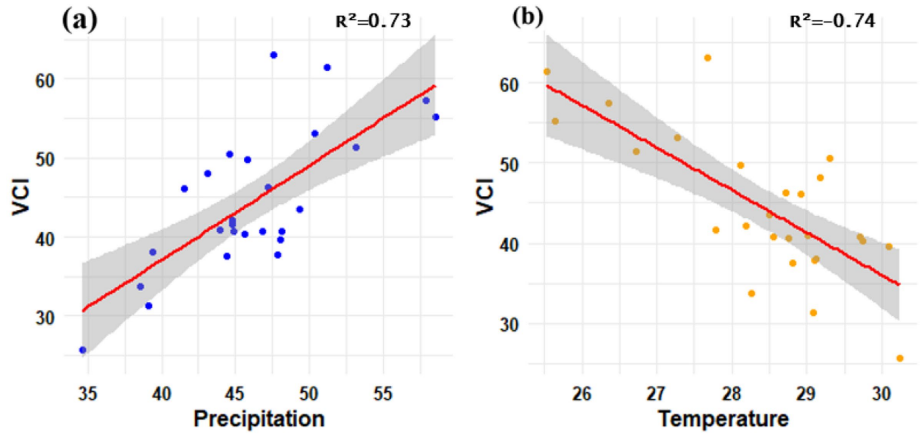


FIGURE 8
Correlation coefficients between growing seasonal VCI and **(a)** precipitation, **(b)** temperature.

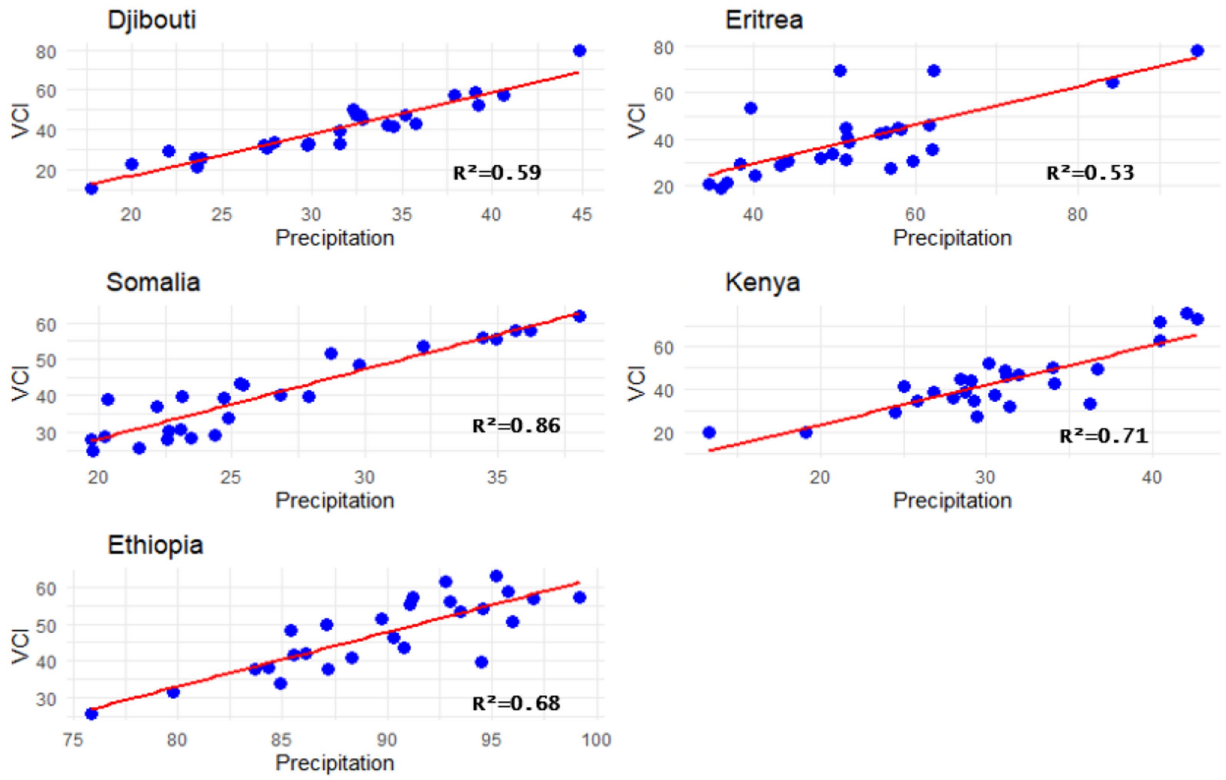


FIGURE 9
Seasonal relationship between VCI and precipitation across each country of the Horn of Africa.

In Kenya, vulnerability is heterogeneous, with predominantly low to moderate values interrupted by areas of higher vulnerability along rain-fed agricultural zones and topographic corridors. These patterns reflect spatial variations in moisture availability, evaporative demand, and vegetation response as captured by the VCI.

Area-based statistics (Figure 11b) show that low and moderate vulnerability categories cover approximately 31 and 24% of the region, respectively, accounting for 55% of the total area. High and very high vulnerability classes represent 22 and 12%, respectively (a combined 34%), while areas of very low vulnerability comprise about 11%. These proportions highlight that while much of the region experiences moderate exposure to drought risk, significant zones of structural vulnerability persist, particularly in western Ethiopia and adjacent highland areas.

3.7 Correlation analysis of seasonal VCI and maize yield

To evaluate the usefulness of the Seasonal Vegetation Condition Index (VCI) as an indicator of drought impacts on crop

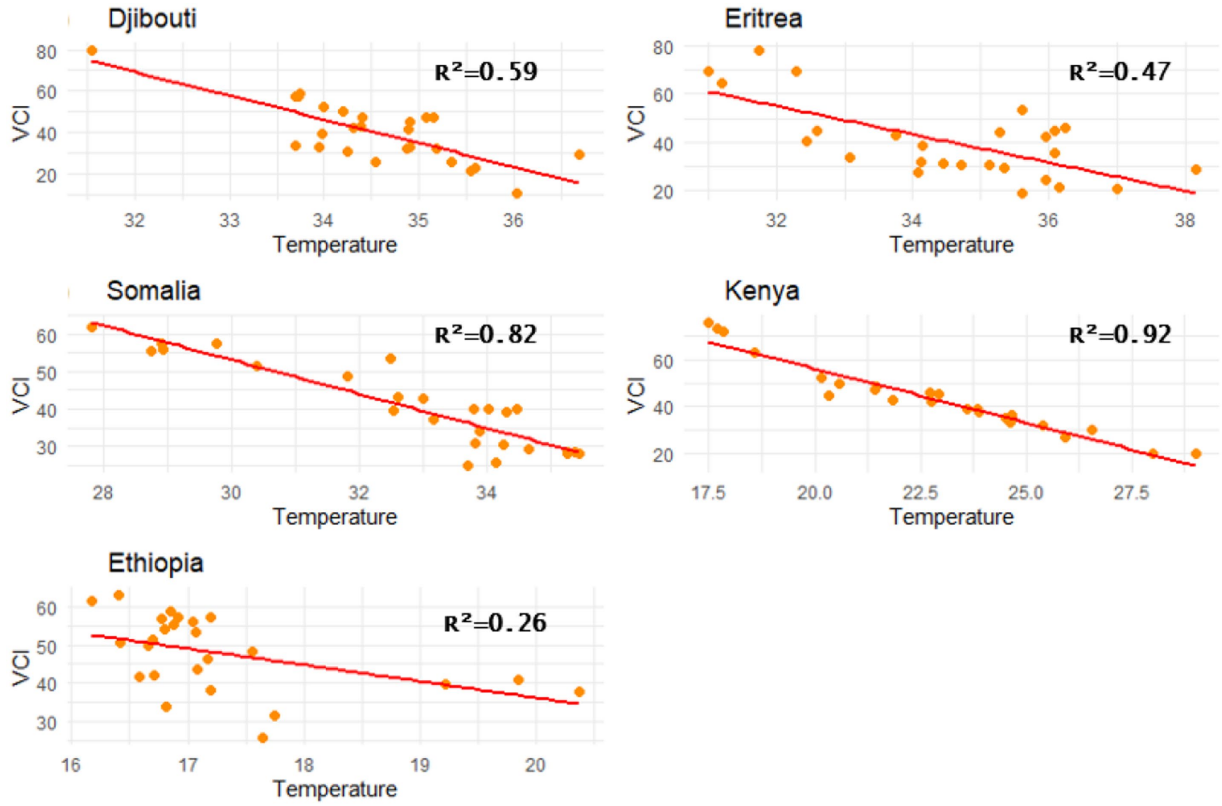


FIGURE 10
Seasonal relationship between VCI and temperature across each country of the Horn of Africa.

performance, we examined its relationship with detrended maize yields for five countries in the Horn of Africa over 1995–2020. Pearson correlation coefficients were calculated between June–August mean VCI and detrended maize yield for each country (Figure 12). The strength of the association varied across countries, but all correlations were positive. Ethiopia showed the strongest relationship ($R = 0.353$, $R^2 = 0.125$, $p = 0.077$), indicating that June–August VCI explained about 13% of the interannual variability in maize yield. Kenya also exhibited a relatively strong positive and statistically significant association ($R = 0.415$, $R^2 = 0.172$, $p = 0.035$). Moderate, statistically significant positive relationships were found for Djibouti ($R = 0.395$, $R^2 = 0.156$, $p = 0.046$) and Eritrea ($R = 0.391$, $R^2 = 0.153$, $p = 0.048$). Somalia showed a weaker, marginally significant positive correlation ($R = 0.335$, $R^2 = 0.112$, $p = 0.094$), with seasonal VCI explaining about 11% of the interannual yield variability.

4 Discussion

4.1 Unveiling patterns and impacts of agricultural drought condition

The results of this study corroborate previous observations regarding the intense drought episodes that have plagued the Horn of Africa (HoA) in recent decades, specifically highlighting the years 2004, 2009, 2011, and 2019. For instance, Alasow et al. (2024) document the severity and impacts of these specific events. However, rather than simply confirming past findings, our analysis reveals a concerning shift in the region's drought dynamics. We observe a transition from more cyclical drought patterns towards persistent and spatially extensive drought events. This trend suggests that interannual variability is increasingly influenced by longer-term climatic shifts, rather than isolated anomalous events, a finding that aligns with the broader understanding of climate change impacts on arid and semi-arid regions, as highlighted in the IPCC's Sixth Assessment Report (Legg, 2021).

The low Vegetation Condition Index (VCI) values (<30%) recorded during these critical drought years underscore the severe vegetation stress, which directly correlates with major food security crises, as reported by Kogan (2018), who demonstrated a strong link between VCI decline and increased food insecurity risk. The analytical value of these findings resides in linking climatic and biophysical stressors—rising temperatures, deficient rainfall, and degraded soils—to ecosystem tipping points where agricultural systems lose functional recovery capacity. The intensification of droughts in Eritrea and Ethiopia reflects the combined impacts of warming, land degradation, and shifts in rainfall seasonality, confirming that recurrent drought is less a discrete hazard and more a symptom of broader hydroclimatic instability, a perspective also emphasized by Sivakumar and Stefanski (2007), in his analysis of land degradation's contribution to increased drought frequency. Importantly, this study advances

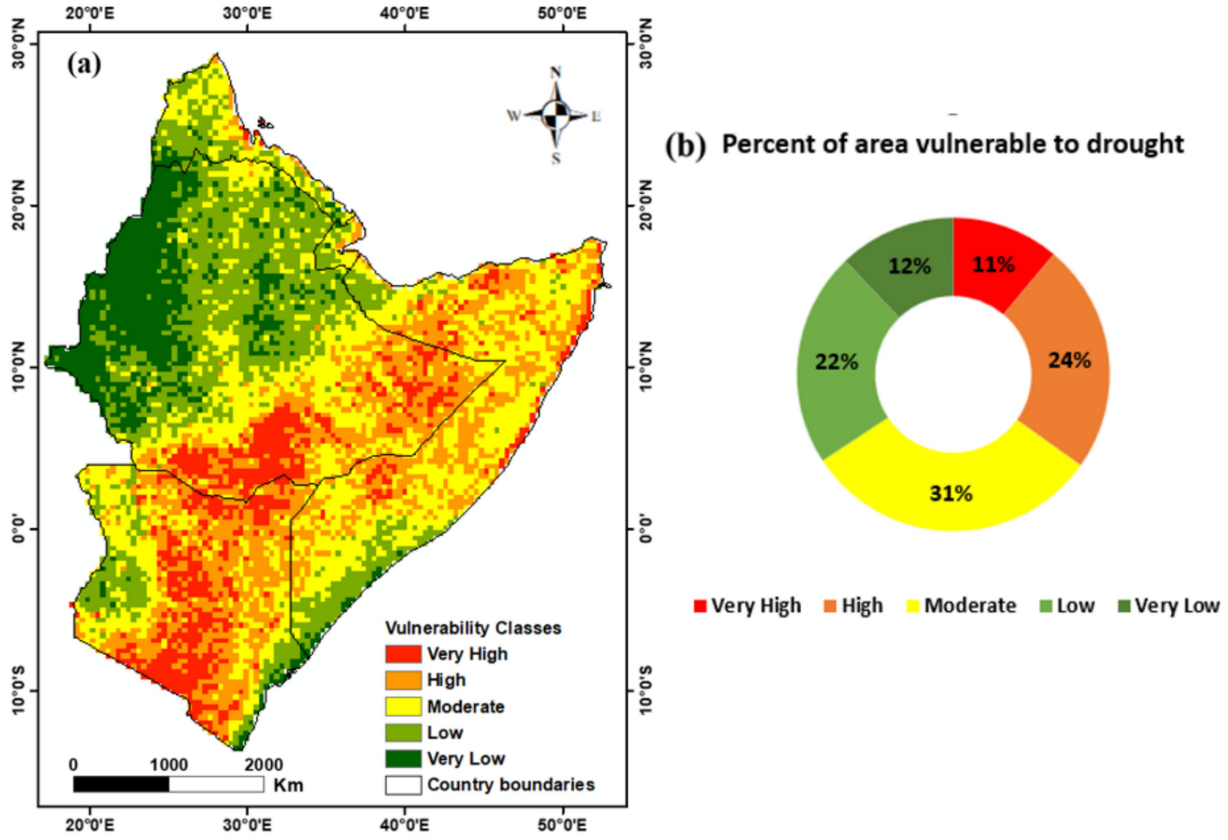


FIGURE 11
Growing season drought vulnerability across the Horn of Africa: **(a)** spatial distribution of vulnerability classes; **(b)** percentage of regional area within each vulnerability class.

existing knowledge by spatially differentiating drought persistence using long-term satellite data, enabling the identification of recurrent hotspots. This spatial understanding is crucial for effective intervention strategies, as exemplified in studies that have used spatial data to optimize drought management and resource allocation (Merabtene et al., 2002; Liu et al., 2021). These insights highlight the urgency of adaptive management that operates both temporally (anticipating drought onset) and spatially (targeting high-risk zones). The implication of these findings points to the need to move beyond reactive responses to drought and invest in proactive strategies that enhance resilience at the local level, incorporating climate-smart agriculture and sustainable land management practices.

4.2 Trends in agricultural drought and climatic variables

Trend analysis using the Mann-Kendall test indicates regionally variable trajectories in vegetation health across the HoA. While overall trends appear statistically stable, sub-regional contrasts are revealing: Djibouti and Eritrea show gradual improvement, whereas Ethiopia exhibits stagnation or mild decline. This divergence underscores how differences in water management, land use, and governance shape local drought outcomes as much as climatic forcing. This is supported by Soomro et al. (2025), which highlights the critical role of effective water resource management in building drought resilience. These findings support the argument that socio-political factors play a key role in mediating the impacts of climate change, as underscored in various studies examining the interplay between governance and climate vulnerability (Kim et al., 2021; Misra et al., 2025). Rather than interpreting stability as resilience, the findings point to compensating dynamics—localized degradation masked by regional averages. Areas exhibiting apparent stability may, in fact, rely on adaptive practices such as small-scale irrigation or soil conservation, which obscure underlying climatic exposure. The results, therefore, suggest that evaluating resilience requires disaggregated, context-specific understanding rather than aggregate regional metrics. This aligns with growing calls for more nuanced, context-specific climate risk assessments, as highlighted by Jones (2019) and Feldmeyer et al. (2020). Integrating these findings with observed precipitation and temperature trends (Figure 7) further illustrates how climatic drivers—particularly enhanced warming and irregular rainfall linked to the Indian Ocean Dipole (IOD)—shape vegetation trends over time. This offers a novel empirical confirmation that vegetative drought in the HoA increasingly reflects compound climate-land interactions rather than single-variable stressors, a point emphasized in research on compound climate events (Zscheischler et al., 2020; Fehlman et al., 2025). This has implications for drought prediction models, which need to account for the complex interplay of climatic and land surface processes to improve accuracy and reliability.

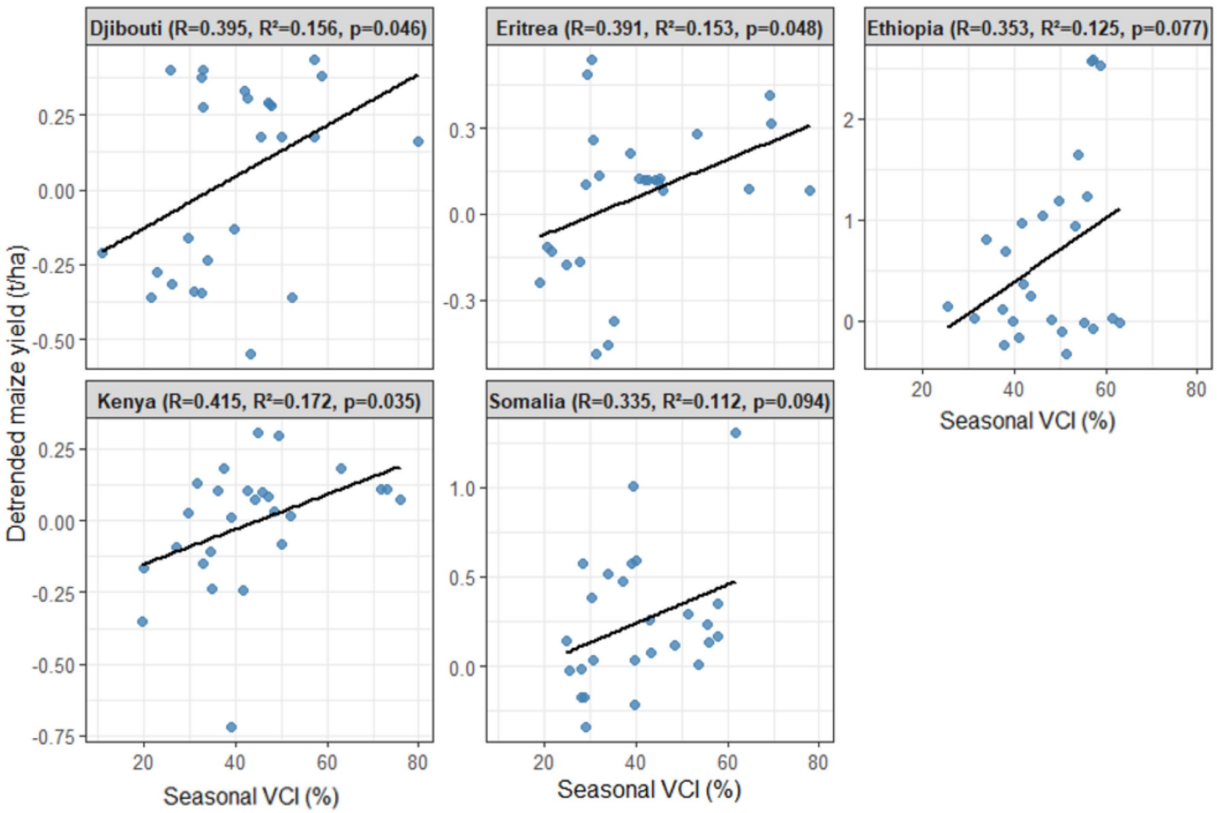


FIGURE 12
Relationship between June–August seasonal Vegetation Condition Index (VCI) and detrended maize yield for five countries in the Horn of Africa.

4.3 Relationship between precipitation, temperature and agricultural drought based VCI

The correlation analysis (Figure 8) reinforces the fundamental control of precipitation on vegetation health ($R^2 = 0.73$) but also reveals important nuance in the role of temperature ($R^2 = -0.74$). While elevated temperatures exacerbate drought through enhanced evapotranspiration and soil moisture loss, their effects can vary by context. For instance, moderate warming in cooler highlands may slightly enhance photosynthesis, whereas in already arid zones it intensifies desiccation stress. This nonlinear response helps explain spatial variability in drought severity across Ethiopia and Kenya. These findings build upon previous research by Chen et al. (2021) and Zhong et al. (2025), highlighting the complex relationship between temperature and vegetation productivity in different climate zones. The high precipitation–VCI correlation in Somalia ($R^2 = 0.86$) and Kenya's strong temperature sensitivity demonstrate that vegetation productivity in the HoA is governed not just by rainfall amount but by its interaction with thermal stress and ecosystem thresholds. These insights extend previous studies by quantifying their joint behavior across multiple national contexts. Understanding this interplay enables more accurate drought monitoring models that incorporate precipitation variability and temperature anomalies rather than relying solely on rainfall metrics. The implications for drought early warning systems are significant. Incorporating temperature sensitivity into drought indices can lead to more timely and accurate warnings, enabling more effective drought preparedness and response measures.

4.4 Spatial heterogeneity in agricultural drought vulnerability

The vulnerability analysis introduces a key contribution: the integration of climate indicators (precipitation, temperature, soil moisture) with vegetation response metrics (VCI) to construct a spatially explicit vulnerability surface. The findings show that while 55% of the HoA falls within low-to-moderate vulnerability categories, large contiguous belts of high-to-very-high vulnerability persist in northwestern Ethiopia and along its transition zones with Eritrea. These areas coincide with regions of intensive rainfed agriculture and high population density—conditions that amplify exposure even when biophysical sensitivity remains constant. This is consistent with evidence that socio-economic conditions intensify vulnerability to climate change, as extensively documented in the literature on climate vulnerability and social inequality (Otto et al., 2017; Li et al., 2023). Conversely, localized pockets of resilience in eastern Ethiopia and Somalia highlight the role of adaptive capacity, such as irrigation infrastructure and livelihood diversification. The interpretation extends beyond mapping vulnerability by linking it to the structural drivers of sensitivity: limited water storage, degraded ecosystems, and socio-economic constraints on adaptive action. The study's cross-country

comparison reveals that vulnerability is not solely a function of climate variability but also institutional capacity and access to adaptive technology. This provides a framework for prioritizing interventions where both environmental pressure and governance limitations intersect. Collectively, these insights position the WOI-based vulnerability framework as a novel contribution to regional drought assessment, offering a replicable approach for tracking compound drought risk in climate-sensitive environments. Strengthening early warning systems and embedding this spatial intelligence into agricultural and disaster risk planning remain critical for building long-term resilience in the Horn of Africa. The policy implications of this research are significant. The vulnerability maps generated can be used to inform targeted interventions, such as investments in water infrastructure, promotion of drought-resistant crops, and strengthening of social safety nets, to reduce the impacts of drought on the most vulnerable populations.

4.5 Limitations and future directions

This study acknowledges several limitations. The accuracy of VCI estimates, derived from remote sensing data, is questioned due to insufficient ground-based validation data (Zhu et al., 2021; Alito and Kerebih, 2024; Bichi et al., 2024). While resampling MODIS data and utilizing CHIRPS precipitation data offer valuable insights, they introduce potential inconsistencies and uncertainties that impact VCI reliability. Furthermore, the inherent biases of index-based maps, focusing on biophysical vulnerability and neglecting socio-economic factors, must be acknowledged.

However, despite these limitations, the significant positive correlations observed between seasonal VCI and detrended maize yields across the Horn of Africa (particularly in Kenya, Djibouti, and Eritrea) suggest VCI's potential as an indicator of drought impacts on crop performance. The relationships observed in Ethiopia and Somalia, while not always statistically significant, further underscore this potential. This is a novel finding, indicating that despite data limitations, VCI can capture meaningful signals related to drought's impact on agriculture in this region.

Future research should prioritize reducing these uncertainties through increased ground-based validation efforts, refined data processing techniques, and integration with socio-economic vulnerability assessments. This would move beyond simply describing the relationship between VCI and yield, allowing for a more holistic and robust agricultural assessment framework capable of informing targeted interventions and building resilience in the Horn of Africa.

5 Conclusion

This study provides a regional assessment of agricultural drought in the Horn of Africa from 1995 to 2020 using the Vegetation Condition Index (VCI) derived from GIMMS NDVI3g and MODIS, linked with precipitation and temperature, and summarized through a Weighted Overlay Index (WOI) of vulnerability. Together, these datasets reveal pronounced interannual variability in vegetation stress and confirm that large areas of the region remain highly exposed to agricultural drought risk.

Rather than a uniform trend toward worsening drought, Mann–Kendall tests indicate a spatially heterogeneous pattern: some areas including parts of Djibouti and Eritrea show relatively stable conditions, while others, notably eastern and central Ethiopia, remain persistently drought-prone. Correlation analyses underscore the dominant influence of climate on vegetation, with precipitation supporting and higher temperatures suppressing VCI, and with June–August greenness moderately associated with detrended maize yields in several countries.

The WOI highlights coherent hotspots of High to Very High agricultural drought vulnerability, particularly in western–northwestern Ethiopia, alongside more moderate and heterogeneous conditions elsewhere, including much of Kenya and Somalia. These results demonstrate the value of integrated, satellite-based monitoring for identifying priority areas and seasons for intervention.

For policymakers, agricultural planners, and humanitarian actors, the findings underscore the need to: (i) target adaptation investments in identified hotspots; (ii) align cropping calendars and management with evolving climate signals; and (iii) couple near-term climate forecasts with operational vegetation monitoring and clear, threshold-based triggers for early action. Strengthening such integrated systems will be critical for reducing the impacts of recurrent agricultural drought on food security and livelihoods in the Horn of Africa.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material; further inquiries can be directed to the corresponding author.

Author contributions

AM: Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft. LL: Writing – review & editing, Software, Visualization, Supervision. JB: Resources, Writing – review & editing, Investigation, Funding acquisition, Supervision, Project administration. IE: Validation, Data curation, Writing – review & editing, Software. VN: Validation, Writing – review & editing, Data curation. MF: Writing – review & editing, Conceptualization, Validation, Data curation. DI: Validation, Writing – review & editing, Data curation. PN: Writing – review & editing, Data curation, Validation. AU: Writing – review & editing, Data curation, Validation, Visualization. FB: Validation, Writing – review & editing, Data curation.

Funding

The author(s) declared that financial support was received for this work and/or its publication. The study was funded and supported by the project of Overseas Science and Education Cooperation Center of the Chinese Academy of Sciences (Grant Number: SAJC202527ZD05).

Acknowledgments

The authors extend their heartfelt thanks to the Alliance of International Science Organizations (ANSO) under the Chinese Academy of Sciences (CAS) for awarding the doctoral scholarship. They also wish to acknowledge the Xinjiang Institute of Ecology and Geography, Chinese Academy of Science, and University of Chinese Academy of Sciences (UCAS), for providing a supportive and conducive environment that facilitated the successful completion of this research project. In addition, the authors are grateful for the opportunity to use the advanced laboratory facilities at these institutions, which significantly contributed to the success of this research endeavor.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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